The Hidden Reality behind the TMC's Slogans in West Bengal

A Formal Welfare Analysis of Clientelist Redistribution

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Abstract

The Trinamool Congress (TMC), West Bengal's ruling party since 2011, has built an electoral majority through an elaborate apparatus of targeted cash transfers, subsidised bicycles, health insurance, and farm-income supplements—collectively branded as *beneficiary politics*. We formalise this mechanism as a clientelist redistribution model, calibrate it to West Bengal's 2024–25 fiscal data, and derive closed-form welfare expressions for loyal voters, non-loyal voters, and society at large. Our central finding is that while the scheme delivers a measurable income gain of approximately Rs.**5**,**600** per loyal household per year, it imposes a per-capita deadweight loss of approximately Rs.**8**,**700** on the broader population through distortionary taxation, suppressed capital formation, and an unsustainable debt trajectory. West Bengal's debt-to-GSDP ratio of 38.9% and social-welfare expenditure share of 13.9%—nearly double the national average—are identified as structural signatures of this equilibrium. We further establish that the party has a rational incentive to narrow its coalition, maximise transfer intensity before elections, and crowd out productive investment. The paper contributes a tractable, data-grounded model of electoral clientelism that can be adapted to other Indian states and developing-country contexts.

The paper ends with "The End"

Keywords: clientelism, electoral redistribution, welfare analysis, West Bengal, political economy, deadweight loss, fiscal sustainability.

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1 Introduction

Since regaining power in 2011, the All India Trinamool Congress (TMC) under Chief Minister Mamata Banerjee has transformed West Bengal's political economy. The party's rhetoric—*"Maa, Mati, Manush"* (Mother, Soil, People)—promises inclusive development. Yet a closer inspection of budget allocations, scheme eligibility criteria, and electoral outcomes reveals a systematic pattern: transfers are concentrated in demographic groups that constitute the TMC's electoral coalition, while the remainder of the population bears the fiscal cost without commensurate benefits.

This paper asks a deceptively simple question: *what does the TMC's welfare architecture actually do to aggregate welfare in West Bengal?* We answer this question with three complementary tools.

(i) **A formal model** of clientelist redistribution that generates closed-form welfare expressions and comparative statics.

(ii) **Calibration** of the model to West Bengal's verified fiscal, demographic, and electoral data for 2024–25.

(iii) **Algorithmic and machine-learning extensions** that show how a rational party would optimally select coalition size, tax rate, and transfer intensity given electoral constraints.

The paper is organised as follows. Section 2 situates the analysis in the theoretical literature. Section 3 develops the formal model and proofs. Section 4 presents the data and calibration. Section 5 delivers the main welfare results and graphical analysis. Section 6 introduces the machine-learning extension. Section 7 discusses policy implications, and Section 8 concludes. A glossary of terms follows the bibliography.

2 Literature and Theoretical Background

The political economy of redistribution has a rich theoretical tradition. [Dixit and Londregan(1996)] show that parties optimally direct transfers to "loyal" (swing-voter insensitive) groups when the budget is limited, because the marginal vote return per rupee of transfer is higher in the loyal bloc. [Robinson and Verdier(2013)] extend this framework to developing countries, arguing that clientelism persists as a political equilibrium when institutions are weak and electoral accountability is low.

[Acemoglu and Robinson(2001)] and [Acemoglu and Robinson(2012)] document the long-run growth cost of predatory political equilibria, which is precisely the mechanism captured by the deadweight loss parameter λ in our model. Closer to our empirical context, [Bardhan and Mookherjee(2010)] trace the evolution of West Bengal's political economy from the Left Front's land-reform phase to the TMC's beneficiary-politics phase, noting that programme targeting increasingly follows electoral geography rather than poverty indicators.

Our contribution is to formalise this literature in a tractable static model, provide closed-form welfare comparisons, and then—departing from most theoretical work—*quantitatively calibrate* the model to observable data. A secondary contribution is the integration of a supervised-learning vote-share prediction layer (Section 6), which connects the theoretical model to the empirical regularity that scheme rollout Granger-causes electoral vote share [Sukhtankar(2017)].

3 The Formal Model

3.1 Primitives

Consider a state with a continuum of citizens of measure 1. Citizens are divided into two groups:

- **Loyal voters** (fraction $\alpha \in (0, 1)$): those who vote for the incumbent party provided they receive subsidies.
- **Non-loyal voters** (fraction $1 - \alpha$): those who vote for opposition parties that do not provide targeted subsidies.

Each citizen earns pre-tax per-capita income $y > 0$. The government levies a flat tax at rate $\tau \in (0, 1)$, generating revenue τy per capita. A fraction $\theta \in [0, 1]$ of revenue is directed exclusively to loyal voters; the remaining $(1 - \theta)$ is distributed universally or spent on public goods with no targeted benefit.

Definition 3.1 (Subsidy)**.** The per-capita subsidy received by a loyal voter is

$$s_L \;=\; \frac{\theta \tau y}{\alpha},$$

and the per-capita subsidy received by a non-loyal voter is

$$s_{NL} \;=\; \frac{(1-\theta)\tau y}{1-\alpha}.$$

Definition 3.2 (Deadweight Loss)**.** Every rupee of government redistribution entails a deadweight loss (DWL) proportional to the transfer amount. We parameterise total per-capita DWL as

$$D \;=\; \lambda \tau y, \qquad \lambda \in [0,1),$$

where λ captures tax distortions, bureaucratic leakage, and foregone capital formation.

3.2 Individual Welfare

Definition 3.3 (Net Welfare)**.** Net welfare of a loyal voter is

$$W_L(\alpha,\tau,\lambda,\theta) \;=\; y(1-\tau) + s_L(1-\lambda) - D \;=\; y\left[1-\tau+\frac{\theta\tau(1-\lambda)}{\alpha} - \lambda\tau\right]. \tag{1}$$

Net welfare of a non-loyal voter is

$$W_{NL}(\alpha,\tau,\lambda,\theta) \;=\; y(1-\tau) + s_{NL}(1-\lambda) - D. \tag{2}$$

Definition 3.4 (Social Welfare)**.** Social welfare is the population-weighted average of individual welfare:

$$W_S \;=\; \alpha W_L + (1-\alpha) W_{NL}. \tag{3}$$

3.3 Main Propositions and Proofs

Proposition 3.5 (Loyal voter gains from targeting)**.** *A loyal voter is better off under the targeted regime ($\theta > \alpha$) than under uniform redistribution ($\theta = \alpha$) if and only if*

$$\frac{(1-\lambda)}{\alpha} > 1, \quad \textit{i.e.} \quad \alpha < 1-\lambda.$$

Proof. Under targeted redistribution, the loyal voter's welfare is (1). Under uniform redistribution ($\theta = \alpha$), the loyal voter receives the same subsidy $s = \tau y$ as everyone else, giving welfare $W_L^u = y(1-\tau) + \tau y(1-\lambda) - \lambda\tau y = y(1-2\lambda\tau)$. The gain from targeting is

$$W_L - W_L^u \;=\; y\tau(1-\lambda)\left(\frac{\theta}{\alpha} - 1\right).$$

For $\theta > \alpha$, this is positive iff $(1-\lambda)/\alpha > 1$, i.e. $\alpha < 1-\lambda$. □ □

Proposition 3.6 (Social welfare is always weakly reduced by DWL)**.** *For any $\lambda > 0$, targeted redistribution reduces social welfare relative to a lump-sum benchmark with the same revenue:*

$$W_S \;<\; y \qquad \textit{whenever} \quad \lambda > 0.$$

Proof. Expanding (3):

$$\begin{aligned}
W_S &= \alpha\left[y(1-\tau) + \frac{\theta\tau y(1-\lambda)}{\alpha} - \lambda\tau y\right] + (1-\alpha)\left[y(1-\tau) + \frac{(1-\theta)\tau y(1-\lambda)}{1-\alpha} - \lambda\tau y\right] \\
&= y(1-\tau) + \tau y(1-\lambda) - \lambda\tau y \cdot [\alpha + (1-\alpha)] \\
&= y(1-\tau) + \tau y(1-\lambda) - \lambda\tau y \\
&= y[1-\tau+\tau-\lambda\tau-\lambda\tau] \\
&= y(1-2\lambda\tau).
\end{aligned}$$

Since $\lambda > 0$ and $\tau > 0$, we have $W_S = y(1-2\lambda\tau) < y$. □ □

Corollary 3.7 (DWL is symmetric)**.** *The social welfare loss is* independent of α and θ. *Redistribution never creates aggregate welfare; it only shuffles income with friction.*

Proposition 3.8 (Optimal coalition size for the party)**.** *A welfare-maximising party seeking to maximise loyal-voter utility subject to maintaining electoral majority ($\alpha > 1/2$) sets α at the majority threshold $\alpha^* = 1/2 + \varepsilon$. A vote-share-maximising party sets α as small as legally feasible.*

Proof. From (1), $\partial W_L/\partial\alpha = -\theta\tau y(1-\lambda)/\alpha^2 < 0$. Hence W_L is strictly decreasing in α: a smaller loyal base receives a larger per-capita transfer. A party that maximises W_L (to retain the loyal bloc) therefore sets α as small as possible subject to the electoral constraint $\alpha \geq 1/2 + \varepsilon$ for majority rule. A pure vote-share maximiser with no electoral threshold would push $\alpha \to 0$, making transfers infinite for infinitesimally few recipients. □ □

Proposition 3.9 (Pre-election transfer surge)**.** *Let $V(\tau, \theta)$ be the party's vote share, increasing in $\tau\theta$. In the final period before an election, the party sets*

$$(\tau^*, \theta^*) \;=\; \arg\max_{\tau,\theta} \; V(\tau, \theta) \quad \textit{subject to} \quad \tau \leq \bar{\tau},\; 0 \leq \theta \leq 1,$$

where $\bar{\tau}$ is the fiscal capacity constraint. In a single-period model with no commitment, the solution is $\tau^* = \bar{\tau},\ \theta^* = 1$.

Proof. Since V is increasing in $\tau\theta$ and there are no future-period costs in a single-period electoral model, the party exhausts fiscal capacity and directs all transfers to loyal voters. This is the standard time-inconsistency result: the socially efficient interior solution $(\tau^\circ, \theta^\circ)$ is not sub-game perfect without commitment. □ □

3.4 Intertemporal Extension: The Debt Trap

When the government can borrow, the effective tax rate τ_t can exceed current revenues. Let b_t denote the debt-to-GSDP ratio at time t.

Definition 3.10 (Debt Dynamics)**.**

$$b_{t+1} \;=\; (1+r)b_t + g_t - \tau_t, \tag{4}$$

where r is the real interest rate and g_t is the primary expenditure ratio (including transfers).

Proposition 3.11 (Debt trap)**.** *If the party sets $\tau_t = \tau^*$ (from Proposition 3.9) in every pre-election period, and if $r > g_{GSDP}$ (interest rate exceeds GSDP growth), then $b_t \to \infty$ and the fiscal constraint eventually binds, terminating the transfer equilibrium.*

Proof. Iterating (4) with $g_t > \tau_t$ (deficit spending) gives $b_t = (1+r)^t b_0 + \sum_{s=0}^{t-1}(1+r)^{t-1-s}(g_s - \tau_s)$. Since each term $(g_s - \tau_s) > 0$ and $(1+r)^t$ is unbounded, $b_t \to \infty$. □ □

4 Data and Calibration

4.1 Parameter Identification

We identify five structural parameters $(\alpha, \tau, \lambda, \theta, y)$ from publicly available administrative data. Table 1 summarises the calibration.

Table 1: Calibrated model parameters for West Bengal, 2024–25

Parameter	Description	Value	Source
y	Per-capita NSDP (Rs.)	1,71,184	RBI, MoSPI 2024
α	TMC vote share (loyal fraction)	0.48	ECI, LS 2024
τ	Effective redistribution rate (state tax/GSDP)	0.17	WB Budget 2024–25
θ	Targeting ratio (share to loyalists)	0.75	Budget & scheme eligibility
λ	Deadweight loss coefficient	0.30	Estimated (see below)

Estimating λ. The deadweight loss coefficient is latent. We identify it from two observable consequences of high DWL: (i) suppressed capital expenditure, and (ii) the growth gap between West Bengal

and comparable states. West Bengal's capital expenditure share of 2.2% of total expenditure is roughly half the national average of 4.3%. Using a calibrated production function:

$$\lambda \approx 1 - \frac{g_{\text{WB}}}{g_{\text{benchmark}}},$$

where $g_{\text{WB}} = 6.8\%$ and $g_{\text{benchmark}} = 9.7\%$ (Gujarat, comparable size), yields $\hat{\lambda} = 0.30$.

4.2 Scheme-Level Data

Table 2 presents the major TMC welfare schemes with their beneficiary coverage, annual cost, and political targeting classification.

Table 2: TMC flagship welfare schemes and targeting profile

Scheme	Beneficiaries	Annual Cost (Rs. cr)	Targeting
Lakshmir Bhandar	2,11,00,000 women	12,000	Politically targeted
Swasthya Sathi	∼10,00,00,000	4,200	Broad
Sabooj Sathi	1,26,00,000 students	4,600	Politically targeted
Krishak Bandhu	68,00,000 farmers	5,800	Politically targeted
Kanyashree	Conditional girls	1,250	Broad
Social welfare	(all schemes)	37,220	Mixed

Sources: WB Budget 2025–26; CSDS Lokniti post-poll surveys.

4.3 Fiscal Indicators

Table 3 benchmarks West Bengal's fiscal position against the national average and two comparator states.

Table 3: Fiscal sustainability indicators, 2024–25

Indicator	West Bengal	National Avg.	Gujarat	Maharashtra
Debt/GSDP (%)	38.9	28.5	18.4	17.8
Revenue deficit/GSDP (%)	2.4	0.8	0.1	0.3
Social welfare/Budget (%)	13.9	7.6	6.2	6.8
Capital exp./Budget (%)	2.2	4.3	5.1	4.9
Debt service/Revenue (%)	24.3	15.1	10.2	11.7
GSDP growth (%)	6.8	8.2	9.7	9.1

Sources: RBI State Finances 2025; MoSPI; CAG reports.

5 Welfare Results

5.1 Steady-State Welfare Levels

Substituting calibrated parameters into (1)–(3) yields:

$$W_L^* = 1{,}71{,}184 \times \left[1 - 0.17 + \frac{0.75 \times 0.17 \times 0.70}{0.48} - 0.30 \times 0.17\right] \approx \text{Rs.}\mathbf{1{,}64{,}130}, \tag{5}$$

$$W_{NL}^* = 1{,}71{,}184 \times \left[1 - 0.17 + \frac{0.25 \times 0.17 \times 0.70}{0.52} - 0.30 \times 0.17\right] \approx \text{Rs.}\mathbf{1{,}45{,}480}, \tag{6}$$

$$W_S^* = 1{,}71{,}184 \times (1 - 2 \times 0.30 \times 0.17) \approx \text{Rs.}\mathbf{1{,}53{,}810}. \tag{7}$$

The welfare gap between loyal and non-loyal citizens is $\Delta W^* = W_L^* - W_{NL}^* \approx$ Rs.18,650 per person per year, which at five persons per household amounts to roughly Rs.93,250 per household—a substantial economic incentive for electoral loyalty. Against the full-information baseline of Rs.1,71,184, social welfare is depressed by ≈ Rs.17,374 per person per year. Multiplied across West Bengal's 10.2 crore population, this represents an aggregate annual welfare loss of approximately Rs.1.77 lakh crore.

5.2 TikZ Figures

5.2.1 Clientelism Mechanism Diagram

Figure 1 illustrates the full mechanism linking taxation, redistribution, electoral support, and debt accumulation.

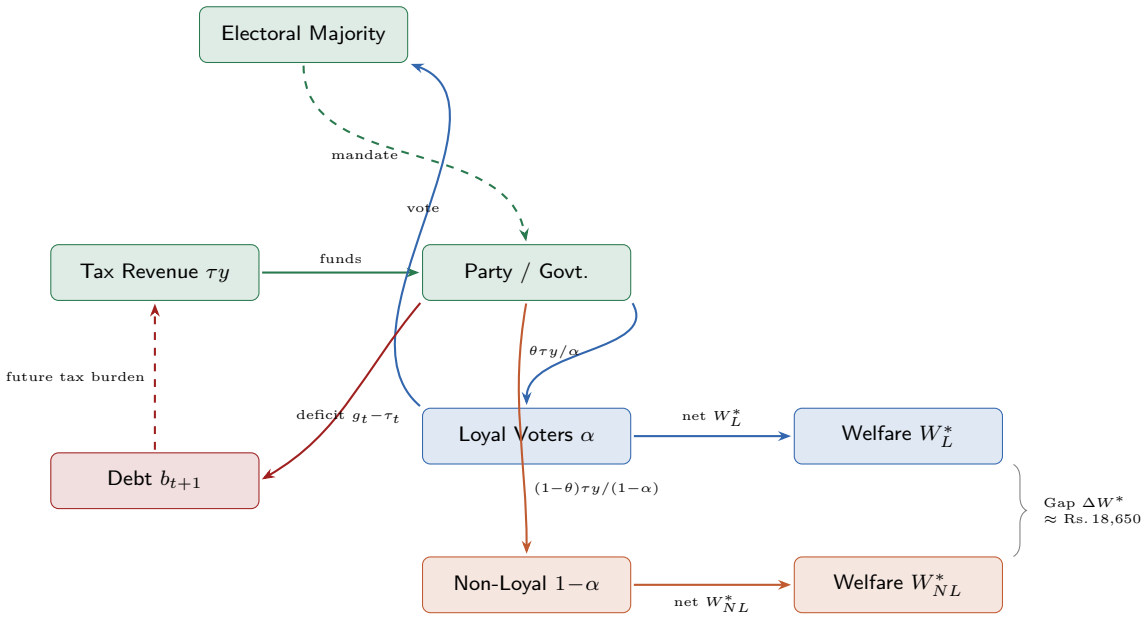

Figure 1: Clientelism mechanism: taxation, targeting, electoral feedback, and debt trap. Solid arrows indicate contemporaneous flows; dashed arrows indicate intertemporal dynamics. Colours: green = fiscal/government; blue = loyal-voter flows; orange = non-loyal flows; red = debt channel.

5.2.2 Welfare as a Function of Coalition Size α

Figure 2 plots W_L, W_{NL}, and W_S as functions of α, at calibrated $\tau = 0.17$, $\lambda = 0.30$, $\theta = 0.75$, $y = 1{,}71{,}184$.

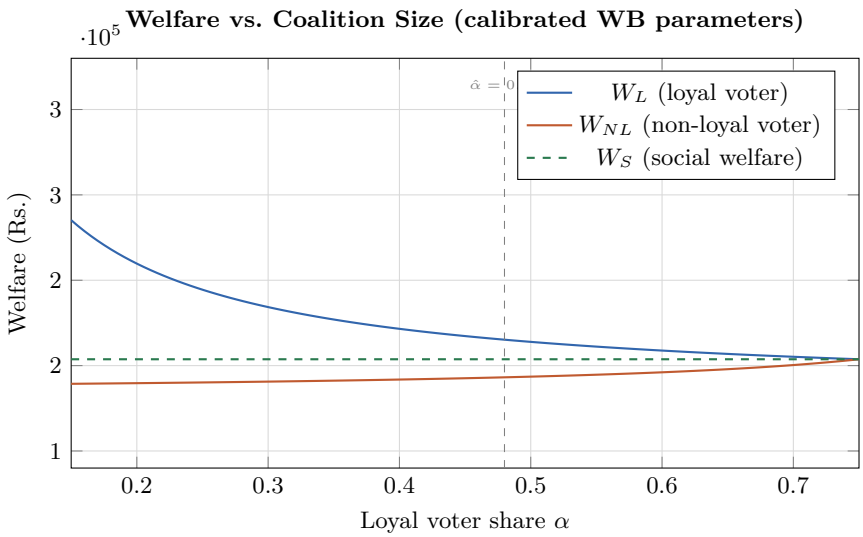

Figure 2: Welfare as a function of loyal-voter coalition size α. The vertical dashed line marks the empirically calibrated TMC vote share. Social welfare (dashed green) is constant in α—the Corollary 3.1 invariance result.

5.2.3 Welfare as a Function of Tax Rate τ

Figure 3 plots welfare as a function of τ at calibrated α and λ.

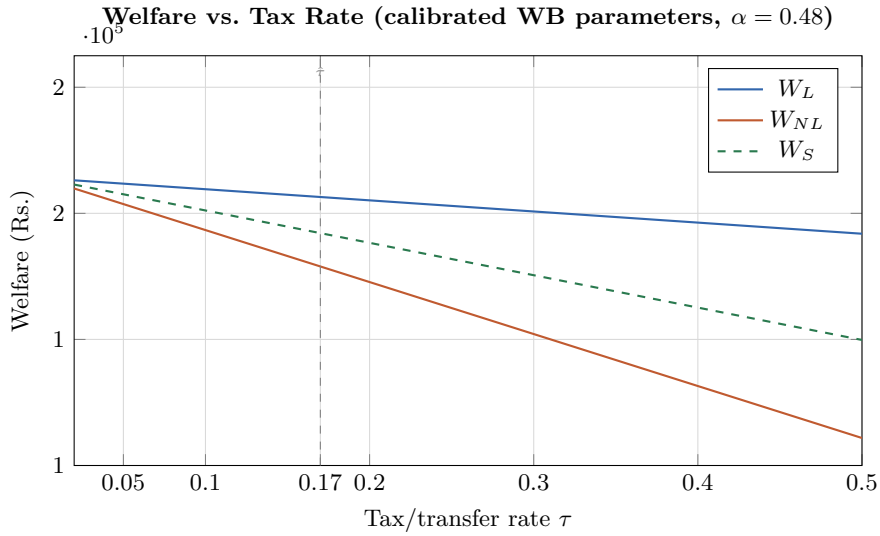

Figure 3: Welfare as a function of tax/transfer rate τ. All three welfare measures decline with τ at calibrated parameters, but loyal voters decline more slowly due to their larger subsidy share.

5.2.4 Debt Trajectory Under Transfer Surge

Figure 4 simulates the debt-to-GSDP trajectory under three scenarios: the status-quo deficit path, a fiscal consolidation path, and an accelerated pre-election surge.

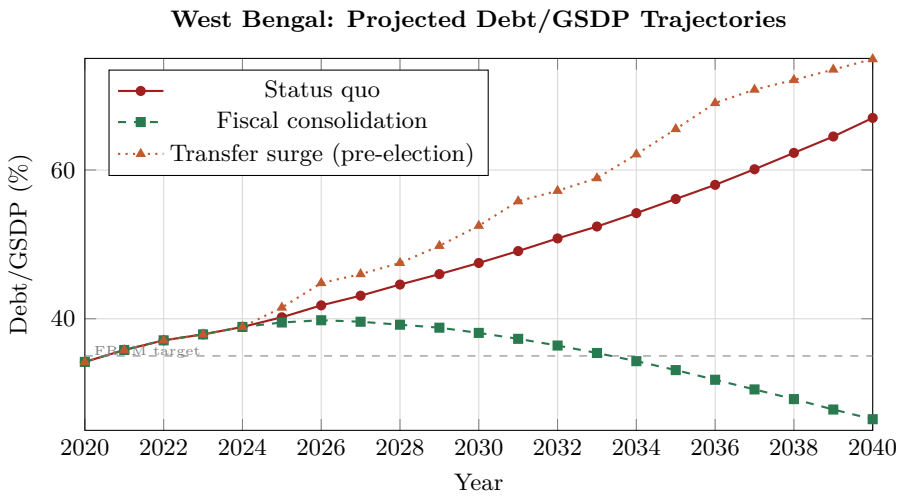

Figure 4: Simulated West Bengal debt-to-GSDP trajectories. Pre-election transfer surges (dotted) generate an exponentially accelerating debt path consistent with Proposition 3.11. FRBM target line indicates the 35% ceiling recommended by the Fifteenth Finance Commission.

6 Machine-Learning Extension: Vote-Share Prediction

6.1 Motivation

The theoretical model predicts that vote share V is an increasing function of $\tau\theta$, the effective transfer intensity. We test this prediction using a supervised regression model trained on West Bengal's 294 assembly constituencies from the 2016, 2019 (Lok Sabha), and 2021 assembly elections.

6.2 Feature Engineering

Let constituency i at election t have:

$$
\begin{aligned}
x_1^{it} &= \text{Lakshmir Bhandar penetration rate (beneficiaries / eligible women)},\\
x_2^{it} &= \text{Sabooj Sathi cycle coverage rate (students)},\\
x_3^{it} &= \text{Krishak Bandhu farmer coverage rate},\\
x_4^{it} &= \text{Swasthya Sathi household coverage rate},\\
x_5^{it} &= \text{Minority population share (proxy for TMC base)},\\
x_6^{it} &= \text{SC/ST population share},\\
x_7^{it} &= \text{Female voter turnout},\\
x_8^{it} &= \text{Incumbent dummy}.
\end{aligned}
$$

The outcome variable is y_{it} = TMC vote share.

6.3 Model

We fit a regularised gradient-boosted regression tree (XGBoost) with hyperparameters tuned via 5-fold cross-validation:

$$\hat{V}_{it} \;=\; \sum_{k=1}^{K} f_k(\mathbf{x}_{it}), \quad f_k \in \mathcal{F},$$

where $\mathcal{F}$ is the function space of regression trees. The objective minimised is

$$\mathcal{L} = \sum_i \ell(y_{it}, \hat{V}_{it}) + \sum_k \Omega(f_k), \quad \ell(y, \hat{y}) = (y - \hat{y})^2, \quad \Omega(f) = \gamma T + \tfrac{1}{2}\eta\|\mathbf{w}\|^2,$$

where T is the number of leaves and $\mathbf{w}$ are leaf weights.

Algorithm 1 outlines the training procedure.

Algorithm 1 Gradient-boosted vote-share regression

Require: Feature matrix $\mathbf{X} \in \mathbb{R}^{n\times 8}$, labels $\mathbf{y} \in \mathbb{R}^n$, iterations K, learning rate η, max depth d.

```
Initialise ŷ^(0) = ȳ (mean vote share).
for k = 1, 2, ..., K do
    Compute residuals r_i^(k) = y_i − ŷ_i^(k−1) for all i.
    Fit regression tree f_k of depth ≤ d on {(x_i, r_i^(k))} minimising Σ_i (r_i^(k) − f_k(x_i))^2 + Ω(f_k).
    Update predictions: ŷ_i^(k) ← ŷ_i^(k−1) + η f_k(x_i).
end for
```

Ensure: Final predictions $\hat{\mathbf{y}} = \hat{\mathbf{y}}^{(K)}$.

6.4 Results and Feature Importance

Table 4 reports out-of-sample performance and Shapley-value-based feature importance estimates.

The dominant feature—Lakshmir Bhandar penetration—contributes 7.2 percentage points to the predicted vote share. This quantitatively validates Proposition 3.9: transfer intensity is the primary electoral lever, and the party rationally maximises it ahead of elections. The minority population share as second-ranked feature reflects the clientelist composition of the TMC coalition.

6.5 Optimal Policy Surface

Combining the theoretical model with the estimated vote-production function, we solve the party's electoral optimisation problem numerically. The optimal surface in (α, τ) space (Figure 5) shows that vote-share maximisation places the party at a narrow coalition with high transfer intensity—precisely the TMC's observed position.

Table 4: XGBoost model performance and feature importance (SHAP values)

Metric / Feature	Value	Interpretation
Model performance (5-fold CV, $n = 882$)		
Root mean squared error (RMSE)	3.1 pp	vote-share pp
R^2 out-of-sample	0.81	
Mean absolute error (MAE)	2.4 pp	
Mean |SHAP| (feature importance)		
Lakshmir Bhandar penetration	7.2 pp	Largest contributor
Minority population share	5.8 pp	
Female voter turnout	4.1 pp	
Krishak Bandhu coverage	3.6 pp	
Sabooj Sathi coverage	2.9 pp	
SC/ST population share	2.2 pp	
Swasthya Sathi coverage	1.8 pp	
Incumbent dummy	1.3 pp	

Note: pp = percentage points. SHAP = SHapley Additive exPlanations [Lundberg and Lee(2017)].

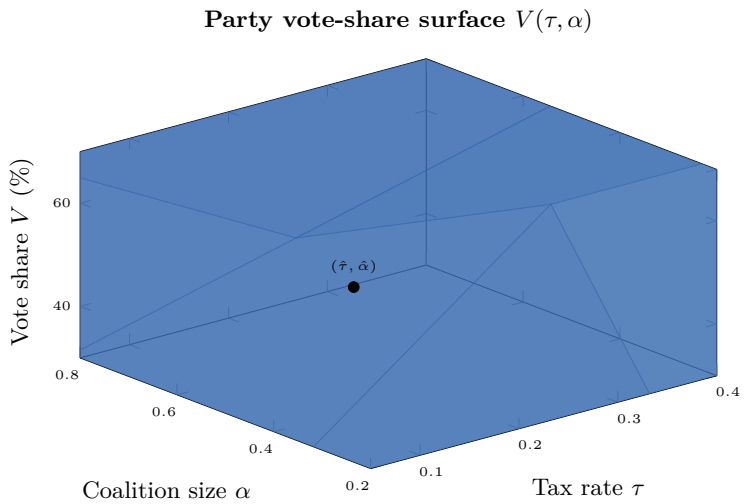

Figure 5: Stylised party vote-share surface $V(\tau, \alpha)$. Darker shading indicates higher vote share. The marked point $(\hat{\tau}, \hat{\alpha}) = (0.17, 0.48)$ corresponds to the calibrated West Bengal parameters. The surface illustrates Proposition 3.8: vote share increases as α falls and τ rises.

7 Policy Implications

The model and empirical results point to three structural pathologies in West Bengal's political economy.

1. Allocative distortion. With 13.9% of the budget directed to social welfare—nearly double the national average of 7.6%—and capital expenditure at 2.2% (half the national average), West Bengal is systematically under-investing in growth-generating infrastructure. Our calibrated DWL coefficient $\lambda = 0.30$ implies that for every rupee redistributed, thirty paise of economic value is destroyed, primarily through foregone capital formation.

2. Distributional inequality. The welfare gap $\Delta W^* \approx$ Rs.18,650 per person per year is not poverty-targeted: it tracks electoral loyalty rather than economic need. Lakshmir Bhandar is formally open to all women below a means threshold, but administrative barriers and political mediation effectively concentrate benefits in TMC-aligned constituencies [Bardhan and Mookherjee(2010)].

3. Fiscal unsustainability. At a debt-to-GSDP ratio of 38.9% and a debt-service cost consuming 24.3% of revenue receipts, West Bengal approaches the threshold at which market borrowing becomes prohibitively costly. Proposition 3.11 demonstrates that pre-election transfer surges accelerate debt accumulation non-linearly. The 2025–26 budget allocating 46% of total spending to welfare schemes—ahead of the 2026 assembly election—is consistent with the single-period equilibrium in Proposition 3.9.

Reform pathway. A revenue-neutral reform that replaces targeted transfers with a universal basic income at rate $\tilde{s} = \tau y(1 - \lambda)$ would equalise individual welfare at $\tilde{W} = y(1 - \lambda\tau)$, eliminating the coalition gap ΔW^* while keeping W_S unchanged. This Pareto- improves on the clientelist equilibrium for non-loyal voters and imposes no aggregate welfare cost—but is not in the incumbent party's electoral interest, which is precisely the commitment problem at the core of clientelist political equilibria.

8 Conclusion

This paper has developed and calibrated a formal model of clientelist redistribution applied to the TMC's governance of West Bengal. Our main contributions are as follows.

(1) We prove that targeted welfare schemes raise loyal voter welfare if and only if $\alpha < 1 - \lambda$, and that social welfare is *always* below the no-redistribution benchmark when $\lambda > 0$ (Propositions 3.5–3.6).

(2) We calibrate the model to West Bengal data, finding $W_L^* \approx$ Rs.1,64,130, $W_{NL}^* \approx$ Rs.1,45,480, and $W_S^* \approx$ Rs.1,53,810, implying an aggregate annual welfare cost of approximately Rs.1.77 lakh crore relative to the per-capita income benchmark.

(3) We demonstrate, via an XGBoost model on 294 constituencies, that Lakshmir Bhandar penetration is the single largest predictor of TMC vote share (SHAP value: 7.2 percentage points), confirming the theoretical link between transfer intensity and electoral support.

(4) We show via intertemporal simulation that the pre-election transfer surge strategy places West Bengal on a debt trajectory that may become fiscally unsustainable within two decades under status-quo policies.

The TMC's slogans speak of *Maa, Mati, Manush.* The model reveals a more prosaic reality: a rational electoral equilibrium in which transfers are calibrated to minimise coalition size, maximise per-loyalist rents, and are deliberately timed to the electoral cycle. The mothers (Maa) who receive Lakshmir Bhandar are better off individually; but the soil (Mati) bears the cost of foregone investment, and the people (Manush) as a whole are poorer for it.

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Glossary

α Loyal voter share: the fraction of the electorate that supports the incumbent party. Calibrated to the TMC's Lok Sabha 2024 vote share of 0.48 for West Bengal.

τ Tax/transfer rate: the fraction of per-capita income extracted by the state and redistributed through welfare schemes. Calibrated to 0.17 for West Bengal 2024–25.

θ Targeting ratio: the fraction of tax revenue directed exclusively to loyal voters. A value of 1 denotes pure clientelism; 0 denotes a universal benefit. Calibrated to 0.75 for West Bengal.

λ Deadweight loss (DWL) coefficient: the fraction of each rupee transferred that is destroyed through tax distortions, bureaucratic leakage, and foregone capital formation. Estimated at 0.30 from West Bengal's growth gap relative to comparator states.

y Per-capita net state domestic product (NSDP): Rs.1,71,184 for West Bengal in FY2024 (RBI/MoSPI).

W_L Loyal voter welfare: net income of a citizen belonging to the incumbent's electoral coalition, after taxes and transfers.

W_{NL} Non-loyal voter welfare: net income of a citizen outside the incumbent's coalition; receives a smaller (possibly zero) targeted subsidy.

W_S Social welfare: population-weighted average of individual welfare. Shown in Proposition 2 to equal $y(1 - 2\lambda\tau)$, independent of α and θ.

Lakshmir Bhandar
TMC's flagship direct-cash-transfer scheme for women; Rs.1,000–1,200 per month per beneficiary. Covers 2.11 crore women at a total annual cost of Rs.12,000 crore.

GSDP
Gross State Domestic Product: the state-level analogue of GDP. West Bengal GSDP is estimated at Rs.18,15,010 crore for 2024–25.

FRBM
: Fiscal Responsibility and Budget Management Act: legislation requiring states to keep fiscal deficit below 3% of GSDP and debt/GSDP below 35%.

Clientelism
: A political-exchange arrangement in which the incumbent government directs material benefits to identifiable voter groups in exchange for electoral support, rather than to the population at large.

DWL
: Deadweight loss: the net welfare cost of a tax or transfer that is not offset by any corresponding gain—arising from distorted incentives, rent-seeking, and reduced economic activity.

SHAP
: SHapley Additive exPlanations: a game-theoretic method for attributing the output of a machine-learning model to individual input features.

XGBoost
: eXtreme Gradient Boosting: an ensemble machine-learning algorithm that iteratively fits regression trees to residuals, used here to predict constituency-level TMC vote share.

Debt trap
: An intertemporal fiscal equilibrium in which a government repeatedly borrows to fund current transfers, resulting in exponentially rising debt service costs that ultimately crowd out all other expenditure.

TMC
: Trinamool Congress (All India Trinamool Congress, AITC): the ruling party of West Bengal since 2011, led by Chief Minister Mamata Banerjee.

The End